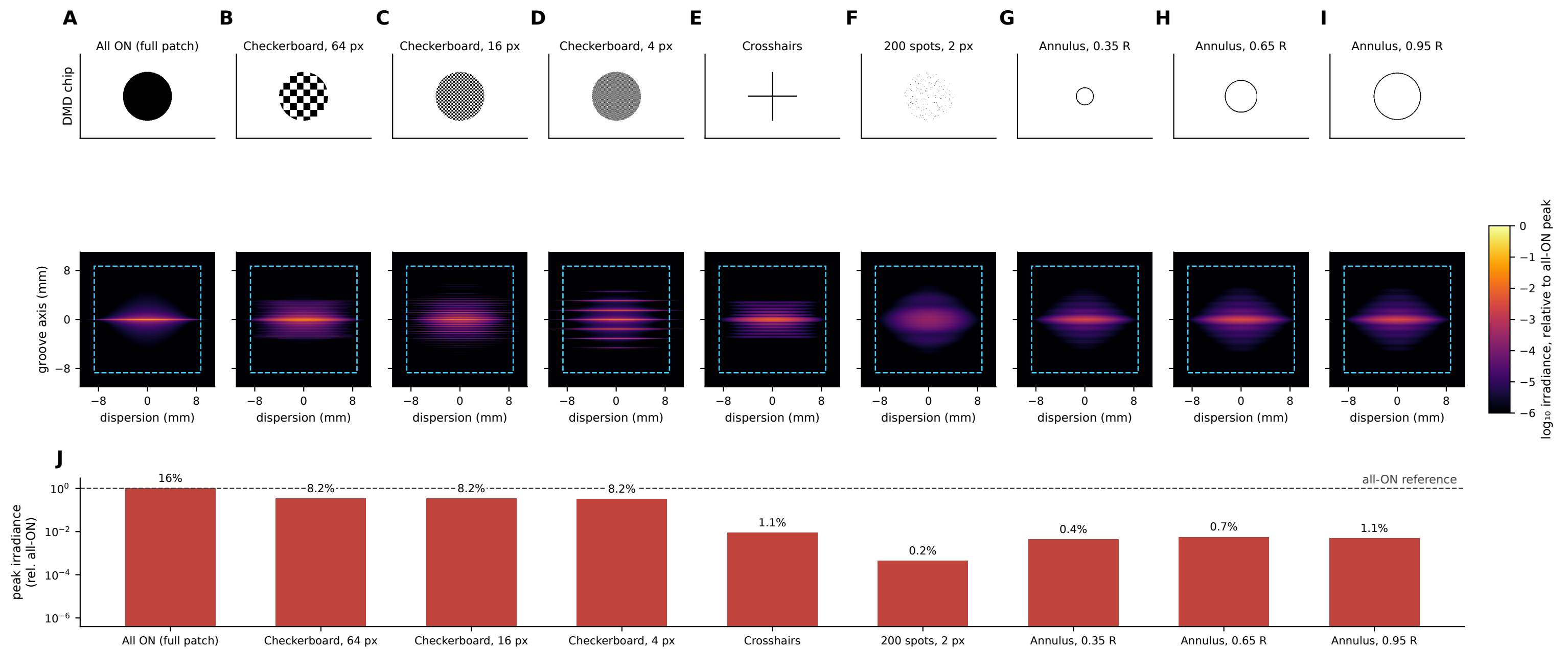




Irradiance at the SLM is the Fourier transform of the DMD pattern, smeared by the laser spectrum. Scalar Fourier propagation of the merged arm to the Meadowlark HSP1K, at the bench-confirmed $f7 = 300$ mm (handoff rev 4 is generated at 250 mm, so its own $f5$ figures are $1.2\times$ larger than these). All constants except $f7$ are parsed live from docs/optics_handoff.md. **(A-I)** Top row, the binary frame written on the DMD, confined to the $\varnothing 5.0$ mm illuminated disc; bottom row, the resulting time-averaged irradiance on the liquid crystal, shown as $\log_{10}$ relative to the all-ON peak over a common 6-decade scale and identical ± 11 mm axes, so panels are directly comparable. Dashed cyan square is the 17.4 mm LC aperture. The horizontal axis is the grating's dispersion direction; because the chip is clocked 45° , each transform is rotated into this lab frame, which is why chip-aligned features (D, E) appear on the diagonals. **(A)** A filled patch puts everything into a single DC order, which the grating then draws out into a spectral line of 0.02×8.6 mm ($1/e^2$, groove $\times$ dispersion). That $\sim 400:1$ smear is what keeps the LC below its damage limit, and why the air-breakdown interlock sits upstream of the grating instead. **(B-D)** Checkerboards at 64, 16 and 4 px place first orders at a radius that grows as $1/\text{period}$, walking light outward from DC. **(E)** Crosshairs transform to a cross rotated 90° from the bars that produced it. **(F)** 200 soma-sized targets, the realistic experiment: a broad speckled envelope with no concentration. **(G-I)** Annuli at 0.35, 0.65 and 0.95 of the patch radius give Bessel-ring transforms whose fringe spacing narrows as the ring grows. **(J)** Peak irradiance per pattern relative to the all-ON frame (log axis, dashed line = that reference); the percentage above each bar is the fraction of the whole chip switched ON, which is the quantity the 50% load-time cap tests. Method: field = pattern $\times$ Gaussian illumination, one FFT to the pupil, rotation into the lab frame, then convolution along dispersion with the 7.7 nm pulse spectrum. The model is unfitted, yet reproduces two independent handoff constants — the 7.2 mm cross-dispersion footprint at $f7 = 250$ and the 0.3604 patch-edge intensity. No pattern clips the aperture (worst case 0.11% of power outside), so the $1.2\times$ footprint growth from $f7 = 250 \rightarrow 300$ does not cost light — but it does spend the margin that made clipping a non-question. The 1.124 anamorphic factor is omitted because the handoff fixes its magnitude but not its sign at the pupil; it would stretch the dispersion axis by $\leq 12\%$ and moves no order across the aperture.